

Tanvir Hossain Ovi

✉ tanvirhossainovi.eee@std.cu.ac.bd 📞 +880 1863 587030

🌐 tanvir-hossain-ovi.me 🔗 LinkedIn 🏠 GitHub 🎓 Google Scholar

EDUCATION

University of Chittagong, Bangladesh

Jan 2020 – Nov 2025

BSc in Electrical and Electronic Engineering | **Engineering Degree with Honours**

CGPA: 3.78 / 4.00 | **Merit Position: 2nd out of 63 students**

Thesis: *EEG-based cognitive load analysis in 360° virtual reality and 2D laptop video modes*

RESEARCH APPOINTMENTS

Research Assistant, Biometric Intelligence and Vision Laboratory

Jul 2025 – Present

Department of Cybersecurity, State University of New York, Canton, USA | Remote

- Preprocessing EEG recordings, designing neural network architectures, feature selection, statistical validation, and drafting manuscripts.
- Supervised by Dr. A M Mahmud Chowdhury, with Dr. Nazmul Siddique of Ulster University, UK, co-supervising the neural signal work.
- First author on the four studies produced during the appointment.

PUBLICATIONS

1. A. Hossain, **T. H. Ovi**, M. A. I. Mahmood, and M. F. Kader, “Classification of Envisioned English Speech from EEG using Deep Learning Approaches,” *Machine Learning with Applications*, vol. 22, p. 100752, 2025.
2. M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, “Impact of Mobile Phone Use on the Brain Activity: Audio Call vs Video Call,” *Acta Psychologica*, vol. 262, p. 106160, 2026.
3. **T. H. Ovi**, M. M. Islam, M. A. Uddin, K. Ghazinour, A. M. M. Chowdhury, and N. Siddique, “Frequency Band Dissociation Between Cognitive State Recognition and Person Identification in EEG,” *Machine Learning with Applications*, p. 100901, 2026.
4. **T. H. Ovi**, A. M. M. Chowdhury, K. Ghazinour, and M. Simsek, “Design of a Clustering-Enabled Multi-Stage Intrusion Detection Framework for APT Mitigation in IoT Networks,” in *Proceedings of the IEEE Conference on Dependable and Secure Computing (DSC)*, 2026. [Accepted]
5. **T. H. Ovi**, T. Forsythe, A. M. M. Chowdhury, N. Siddique, and Y. Li, “Phenotype-aware personalization for cross-session EEG-based emotion recognition,” *Expert Systems With Applications*, 2026. [Under Review]
6. A. Hossain, M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, “Heterogeneous Deep Learning Approaches for EEG-Based Imagined Bangla Word-Level Speech Recognition,” *Computer Speech & Language*, 2026. [Under Review]
7. **T. H. Ovi**, A. M. M. Chowdhury, K. Watanabe, and N. Siddique, “Complementary Raw and Engineered Representations for Focal and Non-Focal Intracranial EEG Classification under a Leakage-

- Controlled Recording-Level Protocol,” *Neurocomputing*, 2026. [Submitted]
8. **T. H. Ovi**, T. F. Arin, M. A. I. Mahmood, and M. F. Kader, “EEG-Based Cognitive Load Analysis in Immersive Virtual Reality and Laptop-Based Video Learning Environments,” *Applied Cognitive Psychology*, 2026. [Submitted]
 9. T. F. Arin, **T. H. Ovi**, and M. F. Kader, “Bangla Imagined Speech Using EEG to Understand Emotional Words,” *Acta Psychologica*, 2026. [Submitted]

RESEARCH EXPERIENCE

Cross-Subject Alignment of Image-Evoked EEG with CLIP Visual Embeddings 2026–present

- Aligning image-evoked EEG responses from the THINGS-EEG dataset with CLIP visual embeddings to test how much of a shared image representation survives in scalp recordings.
- Found the alignment stable within a participant and unstable across participants, which places the limit on the subject-specific mapping rather than on the shared representation.
- Testing whether that subject-specific component can be predicted from a short recording instead of being fitted separately for every participant.

Complementary Raw and Engineered Representations for Focal and Non-Focal iEEG Classification 2026

- Traced the inflated accuracies in prior Bern-Barcelona benchmark work to window-level splits, then rebuilt the partition at the recording level over all 7,500 recordings and repeated every condition across three training seeds.
- Proposed CSRA-EF Net, which pairs a shared cross-channel raw-signal encoder with a 280-dimensional engineered-feature branch (differential entropy, spectral entropy, Hjorth parameters, inter-channel correlation) through token cross-attention.
- Reached 96.27% accuracy and 0.994 AUC, above the strongest raw baseline by 4.71 and the strongest feature-only baseline by 6.04 percentage points.
- Showed by ablation, under a pre-declared one-point equivalence margin, that the engineered-feature branch is the only component with a detectable effect, and that no single frequency band beats a random 43-descriptor subset.

Phenotype-Aware Personalization for Cross-Session EEG Emotion Recognition 2025–2026

- Analyzed the SEED-IV dataset across three weekly sessions per participant to test whether cross-session decline in EEG emotion recognition is random session drift or structured individual variation.
- Partitioned performance variance by two-way ANOVA and placed 80.1% of the explained variance in the participant-by-emotion interaction, then grouped participants into phenotypes by hierarchical clustering of those profiles.
- Raised cross-session accuracy from 39.96% to 88.71% by drawing calibration data from inside a participant’s own phenotype instead of applying a uniform correction, and to 91.92% with an exploratory ensemble.

EEG-Based Frequency Band Analysis for Cognitive State and Biometric Applications 2025

- Independently analyzed a public EEG dataset of 20 participants recorded across three weekly sessions, asking whether the frequency band that best classifies a task is also the band that best identifies a person.
- Built the preprocessing pipeline with ICA artifact removal and evaluated EEGNet with squeeze-and-excitation blocks, channel-mixing variants, and EEGPT across seven frequency bands.
- Found delta reached 86.9% task classification and only 15.6 to 23.6% authentication, while broad-band reached 69.9% authentication, with task and authentication accuracy significantly negatively correlated.

Cognitive Load Analysis in Virtual Reality and Traditional Display Environments 2024–2025

- Designed and ran the protocol comparing cognitive load in 360 degree immersive VR against a 2D laptop display, recording 30 participants on a 14 channel Emotiv EPOC X.
- Built the preprocessing pipeline and the theta to alpha ratio, engagement index and beta to alpha ratio measures, validated by MANOVA and ANOVA with FDR correction.
- Found VR lowered cognitive load by 21% and raised engagement by 59%, with 83.5% classification accuracy across SVM, Random Forest and kNN under five-fold cross validation and the theta to alpha ratio as the strongest marker.

HONORS AND AWARDS

- First Runner-up, Student Merit Award, EEE Fest, University of Chittagong *Feb 2025*
- Champion, Robo Soccer Competition, EEE Fest, University of Chittagong *Jan 2023*
- Second Runner-up, Robo Soccer Competition, Engineering Day, University of Chittagong *Jun 2022*
- Principal's Award for Academic Excellence, Dhaka Imperial College, for 4th position in the Summer Semester Final Examination, Science Group *Sep 2018*
- Government Junior Scholarship, awarded on the Junior School Certificate (JSC) examination results, Chatkhil Panch Gaon Government High School *2014*
- Industrial Training, General Electric Manufacturing Company Limited (14 days) *Jun-Jul 2024*

STANDARDISED TESTS

IELTS Academic *14 April 2026*
 Overall Band Score: **7.5** | Listening: 8.0 Reading: 8.0 Writing: 6.0 Speaking: 7.0

TECHNICAL SKILLS

- **Programming & Frameworks:** Python (3 yrs), TensorFlow, Keras, PyTorch, Scikit-learn (Machine Learning & Deep Learning: 2 yrs)
- **Signal Processing:** MNE-Python, EEGLAB, EEG preprocessing and feature extraction
- **Statistical Analysis:** Hypothesis testing (t-test, Wilcoxon, ANOVA, MANOVA), data modeling
- **Computer Vision:** MediaPipe, OpenCV, ROI extraction, image standardization (NFIQ2)
- **Data Visualization:** Matplotlib, Seaborn, Tableau

- **Tools & Platforms:** LaTeX, Google Colab, Git, Linux
- **Simulation:** Advanced Design System (ADS), RSoft, Pspice

REFERENCES

Dr. Nazmul Siddique

Senior Lecturer (Associate Professor), School of Computing, Engineering and Intelligent Systems
Ulster University, Londonderry
Relation: Research Advisor | nh.siddique@ulster.ac.uk

Dr. A M Mahmud Chowdhury

Assistant Professor, Department of Cybersecurity
State University of New York, Canton
Relation: Research Advisor | Chowdhuryam@canton.edu

Dr. Kazi Tanvir Ahmmed

Professor & Chairman, Department of Electrical and Electronic Engineering
University of Chittagong, Chattogram
Relation: Academic Advisor | tanvir@cu.ac.bd